VDA9.1

Hugo Lladró Prats

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Exercise 1 (Introduction to ParaView, 13 Points)

a) Load the file headsq.vti into a ParaView pipeline. What is the grid type of this dataset? What is the type and range of its values? (2P)

b) A basic and widely used way to visualize 3D CT data is via 2D slices. Different mappings from Hounsfield units to grayscale are used depending on the structures of interest. To examine brain, bone, or potential intracranial hemorrhage, common ranges ("windows") are [0, 80], [770, 1730] and [85, 265] Hounsfield units, respectively. Create three corresponding images, which should look similar to the ones in Figure 1. (6P)
Hint: The values in the given input file correspond to shifted Hounsfield units: 1024 has been added in order to store them as unsigned values.

c) Also create a 3D visualization showing two nested surfaces as in the left part of Figure 1: The outer surface should show the skin, and it should be rendered in a different color as well as semi-transparently in order to reveal an inner surface, which should visualize the bones. (5P)
Hint: This can be achieved with isosurfaces, which are available via ParaView's Contour filter.

a)

We can find this data in the "information" tab. The grid is a tridimensional uniform rectilinear grid. The values are unsigned short integers with values in the range [0, 4095].

b)

To get the original HU we have to add 1024 to the shown ranges. We create a slice and use a greyscale palette with the given values to obtain the following figures

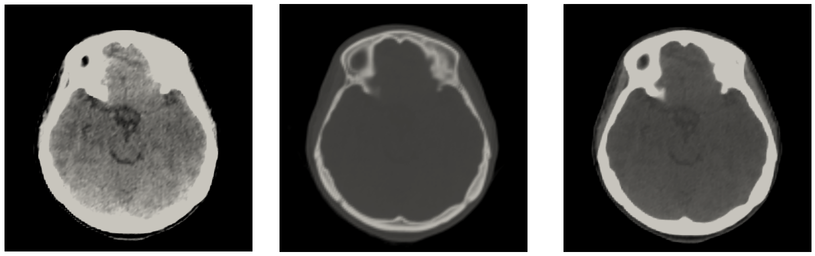

Figure 1: Different greyscale mappings of the data centering on brain damage, bone damage, or potential intracranial hemorrhage respectively.

c)

By using a default transfer function, we estimate that the skin can be found in the range [900, 910], and the bones in the range [1124, 1224]. Using the visualization that the exercise suggests we get the following picture.

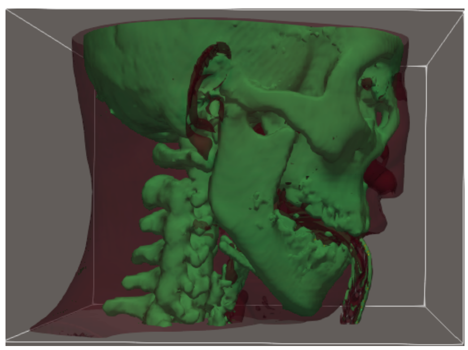

Figure 2: Figure that shows the skin and bones in different layers